

Which Way Does the Condition Point?

AP Statistics · Unit 2 · Topic 2.6 · TEACHER KEY

CED TETHER

- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **Enduring Understanding VAR-4** — The likelihood of a random event can be quantified.
- **LO VAR-4.D** — Calculate conditional probabilities. [Skill 3.A]
- **EK VAR-4.D.1** — The probability that event A will occur given that event B has occurred is called a conditional probability and denoted $P(A | B) = P(A \cap B) / P(B)$.
- **EK VAR-4.D.2** — The multiplication rule states that the probability that events A and B both will occur is equal to the probability that event A will occur multiplied by the probability that event B will occur, given that A has occurred. This is denoted $P(A \cap B) = P(A) \cdot P(B | A)$.

Essential question: Why can reversing the condition produce a very different probability, even when both probabilities use the same table?

DOK-3 skill: 4.B · **FRQ pattern:** conditional-direction-confusion-p-a-given-b-vs-b-given-a

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Rejects Mina using 45/140 or about 32.1 percent
- **reasoning-2** (required) — Uses 45 with the condition plus 95 without to form the positive group
- **reasoning-3** (required) — Explains the change in conditioning direction and denominator

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Using the sensitivity $45/50$ as the chance of condition after a positive result
- Dividing 45 by all 1,000 screened students instead of by the 140 positive tests
- Calling the 95 positives among students without the condition true positives
- Ignoring the 5 percent prevalence when explaining why the two conditional probabilities differ

[STAR] THE PROBLEM — annotated

Suppose a school screens 1,000 students for a hypothetical condition. Exactly 5% have the condition. The test has 90% sensitivity: among students with the condition, 90% test positive. Its false-positive rate is 10%: among students without the condition, 10% test positive. The implied counts are shown.

Mina: “The test catches 90% of students who have the condition, so a positive result means a student has a 90% chance of having it.”

Luis: “We need to count everyone who tested positive before finding the chance of having the condition.”

Screening results (counts)

Status	Pos.	Neg.	Total
Condition	45	5	50
No condition	95	855	950
Total	140	860	1,000

FIRST TAKE — one sentence before you work the parts

Does a positive result mean a 90% chance of having the condition? Choose a claim and cite one table count.

Key: Not graded. The familiar 90 percent number is intentionally tempting; the revision should use the full positive-test group.

[BOOK] THE GIVEN EVENT SETS THE DENOMINATOR (keep on the page)

$P(A \mid B)$ means the probability of A **given** B . Restrict attention to the outcomes in B : $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\text{count in both } A \text{ and } B}{\text{count in } B}$. In general, $P(A \mid B) \neq P(B \mid A)$ because the denominators are different. **Sensitivity** is $P(\text{positive} \mid \text{condition})$. The **false-positive rate** is $P(\text{positive} \mid \text{no condition})$.

DOK 1 (a) For $P(\text{condition} \mid \text{positive})$, identify the numerator and denominator counts.

Key: The numerator is 45 students who both have the condition and tested positive. The denominator is all $45 + 95 = 140$ students who tested positive.

DOK 2 (b) Find $P(\text{positive} \mid \text{condition})$ and $P(\text{condition} \mid \text{positive})$. Show each count fraction and percent.

Key: $P(\text{positive} \mid \text{condition}) = 45/50 = 0.90 = 90\%$. In the other direction, $P(\text{condition} \mid \text{positive}) = 45/140 = 9/28 \approx 0.321 = 32.1\%$.

DOK 3 (c) Is Mina’s claim supported? Use the positive-test counts for students with and without the condition. Explain how reversing given changes the group in the denominator. Write 3–4 sentences.

Key: Luis is right: Mina’s 90% uses the 50 students with the condition as the denominator. After a positive result, the relevant group is the 140 students who tested positive: 45 have the condition and 95 do not. Thus the chance of the condition given positive is $45/140 \approx 32.1\%$, not 90%. Reversing the given event changes the denominator and answers a different question.